AKHIL KUMAR

DevOps & Cloud Infrastructure Engineer

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PROFESSIONAL SUMMARY

DevOps and cloud infrastructure engineer focused on AWS, Terraform, Kubernetes, and CI/CD. Hands-on portfolio work spans event-driven IoT infrastructure, EKS, ECS Fargate, SQS/DLQs, Redis, PostgreSQL, observability, autoscaling, and GitHub Actions with AWS IAM OIDC. Brings prior full-stack production experience and a strong focus on failure modes, infrastructure automation, and operational reliability.

TECHNICAL SKILLS

Cloud & IaC: AWS (IoT Core, ECS Fargate, EKS, SQS, DynamoDB, Step Functions, ECR, SSM, CloudWatch, SNS, RDS, EC2, S3, ALB, Route 53, ACM) · Terraform

Containers & Orchestration: Docker · Kubernetes · AWS EKS · Helm · eksctl

CI/CD & DevOps: GitHub Actions · Jenkins · Git · SonarQube · Trivy · OIDC/IAM

Monitoring: Prometheus · Grafana · CloudWatch · Structured JSON Logging

OS & Scripting: Linux (Ubuntu) · Bash · Python

Development & Data: FastAPI · JavaScript · Node.js · React · Express · PostgreSQL · Redis · SQL · MongoDB

PROJECTS

Smart Helmet IoT Edge Infrastructure (AWS) | May 2026 - Present

Stack: AWS IoT Core · ECS Fargate · Step Functions · SQS · DynamoDB · ECR · SSM · CloudWatch · SNS · Terraform · GitHub Actions/OIDC | github.com/akhilkumar0024/IOT-EDGE-INFRA-SmartHelmet-SIM

- Designed end-to-end event flows covering 11 major flows and 13 failure/configuration branches across device, cloud, CI/CD, and operational scenarios; translated the core architecture into modular Terraform infrastructure.
- Used Python-based MQTT simulator and load-test tooling for normal telemetry, crash, false-positive, override, and LWT events; exercised the ingestion path with a 10,000-message MQTT load test.
- Implemented AWS IoT Core → SQS routing with six dedicated queues and DLQs for telemetry, crash, alert, override, LWT, and control events; ECS Fargate consumers persist telemetry and process crash/alert events through DynamoDB and Step Functions.
- Implemented Step Functions alert countdown and reconciliation workflows with a DynamoDB execution registry, rider cancellation/override handling, and emergency notification paths.
- Implemented CloudWatch/SNS alerting for DLQ backlogs and DynamoDB throttling, plus ECS target-tracking autoscaling driven by SQS queue depth.
- Implemented GitHub Actions CI/CD with OIDC-based IAM roles for Terraform PR plan, main-branch apply, and application image deployment to ECR.

Cloud-Native URL Shortener & Analytics Platform | Sep 2026 - Present

Stack: AWS EKS · Terraform · VPC · SQS/DLQ · Redis · PostgreSQL · FastAPI · Docker · Prometheus · React · Nginx | github.com/akhilkumar0024/url-shortener-eks-platform

- Built a distributed URL-shortening platform with separate API, analytics backend, and frontend services, using Redis caching with PostgreSQL fallback for URL resolution.
- Implemented asynchronous click-event processing with AWS SQS, capturing request metadata and configuring a dead-letter queue and redrive policy for failed message handling.
- Provisioned AWS infrastructure with modular Terraform, including VPC, multi-AZ public/private subnets, Internet/NAT gateways, EKS cluster, managed node group, IAM roles, OIDC provider, and SQS queues.
- Containerized services with Docker/Docker Compose, including multi-stage React/Nginx builds and isolated Alembic migration jobs for service-owned database schemas.
- Added operational observability through Prometheus request/latency metrics, Kubernetes health endpoints, and structured JSON logging.

BookMyShow Clone - DevOps Capstone | Mar 2026 - Apr 2026

Stack: Jenkins · Docker · AWS EKS · eksctl · SonarQube · Trivy · Prometheus · Grafana · Helm | github.com/akhilkumar0024/Book-My-Show

- Built a CI/CD pipeline covering build, static analysis, container vulnerability scanning, image push, and EKS deployment.
- Deployed Prometheus and Grafana in EKS via Helm and validated the application progressively from Docker deployment to Kubernetes on EKS.

EXPERIENCE

Full Stack Developer - Kapiva (Adret Retail Pvt. Ltd.) | Jul 2025 - Nov 2025

Stack: MongoDB · Express · React · Node.js | Bengaluru, Karnataka

- Designed and delivered an audit-logging system for an internal operations dashboard, providing traceability of user actions.
- Built and shipped a clinical workflow feature for an internal Doctors Workbench supporting prescriptions, patient histories, health-device data ingestion, and vital visualisation.
- Delivered features end-to-end in a production B2C environment, from implementation through integration and release.

EDUCATION & TRAINING

DevOps & Cloud Engineering Program - StarAgile | Nov 2025 - Feb 2026 | Jenkins, Docker, Kubernetes, Terraform, Ansible, AWS, Prometheus, Grafana

PG-DAC - CDAC, Ministry of Electronics & IT, Govt. of India | Sep 2023 - Feb 2024 | Advanced Computing

Full Stack Developer Program - Masai School | Nov 2024 - Jun 2025 | JavaScript, React, Node.js, Databases, System Design

B.Tech - Computer Science & Engineering, College of Engineering and Management, Punnapra | 2015 - 2019 | APJ Abdul Kalam Technological University